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Part 2

Transformations and Solving Equations Test

Answer the following questions. Show the movements required to transform your graphs as we have done in class. Do not use a graphing calculator. This test is due on Tuesday, November 25th. Do not work with anyone or go to the Math Lab for assistance. Doing so will result in earning a 0 on this test.

- 1) Using the graph to the right, sketch the left-hand side and the right-hand side of this equation and determine the solution graphically.

$$2x + 1 = -2x - 3$$

$mx + b \quad mx + b$

- 2) Using algebra, solve the above equation to determine the solution.

$$2x + 1 = -2x - 3$$

$-1 \quad -1$

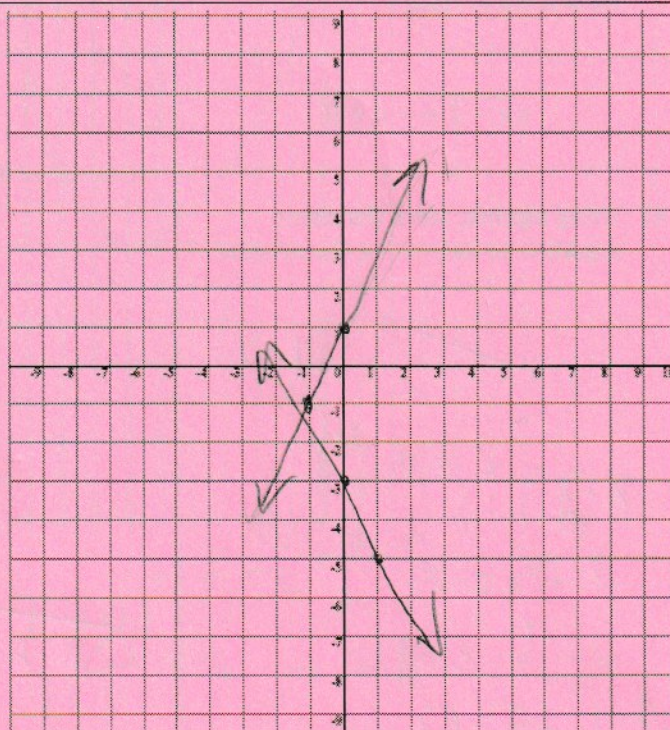
$$2x = -2x - 4$$

$+2x \quad +2x$

$$4x = -4$$

$\div 4 \quad \div 4$

$$x = -1$$



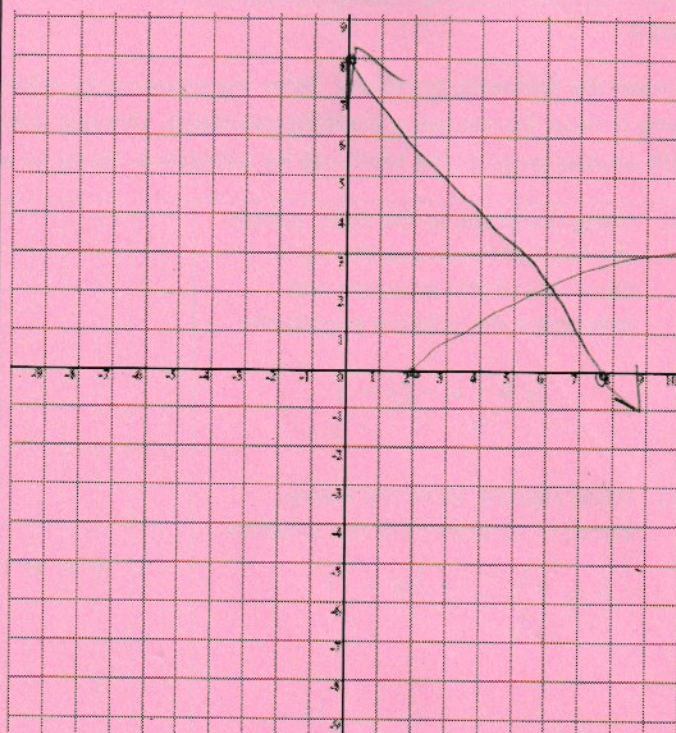
$$(-x + 8)(-x + 8)$$

$$x^2 - 8x + 8x - 64$$

- 3) Using the graph to the right, sketch the left-hand side and the right-hand side of this equation and determine the solution graphically. Show your movements of the parent function for each equation as we have done in class.

$$\sqrt{x-2} = -x+8$$

- 4) Using algebra, solve the above equation to determine the solution.



$$\sqrt{x-2} = (-x+8)$$

$$x-2 = x^2 - 16x + 64$$

$$x^2 - 17x + 66 = 0$$

$$\sqrt{11-2} = -11+8 \text{ or } \sqrt{6-2} = -6+8$$

$$\sqrt{9} = -4 \text{ or } \sqrt{4} = 2$$

$$2 = 2$$

$$(x-11)(x-6) = 0$$

$$x-11 = 0 \text{ or } x-6 = 0$$

$$x \neq 11 \text{ or } x = 6$$

- 5) Using the graph to the right, sketch the left-hand side and the right-hand side of this equation and determine the solution graphically. Show your movements of the parent function for each equation as we have done in class.

$$2(x+2)^2 - 2 = 6$$

- 6) Using algebra, solve the above equation to determine the solution.

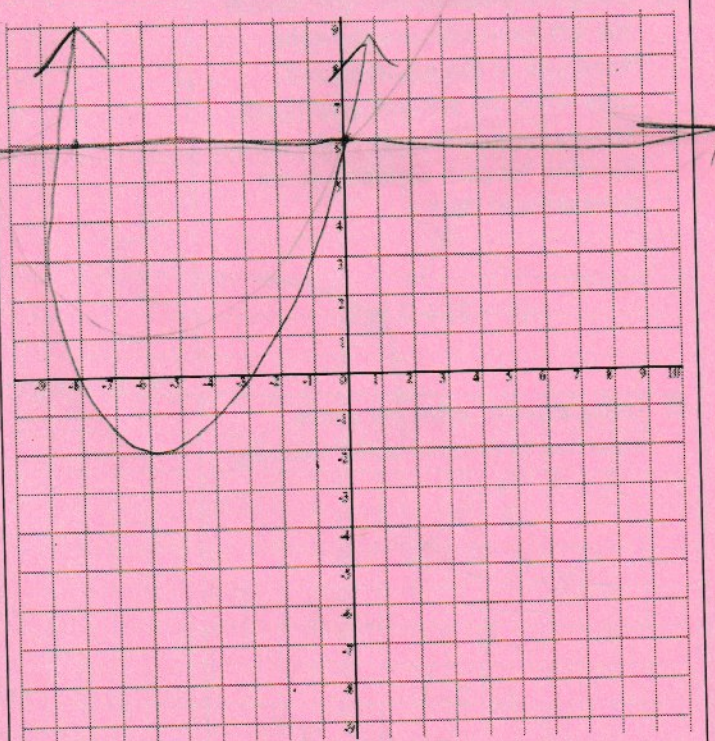
$$2(x+2)^2 - 2 = 6$$

$$\frac{2(x+2)^2}{2} = \frac{8}{2}$$

$$\sqrt{(x+2)^2} = \sqrt{4}$$

$$x+2 = 2 \quad x+2 = -2$$

$$x = 0, x = -4$$



$$2(0+2)^2 - 2 = 6 \quad \text{or} \quad 2(4+2)^2 - 2 = 6$$

$$2(2)^2 - 2 = 6 \quad \text{or} \quad 2(6)^2 - 2 = 6$$

$$2(4)^2 - 2 = 6 \quad \text{or} \quad 2(36)^2 - 2 = 6$$

$$8 - 2 = 6 \quad \text{or} \quad 12 - 2 = 6$$

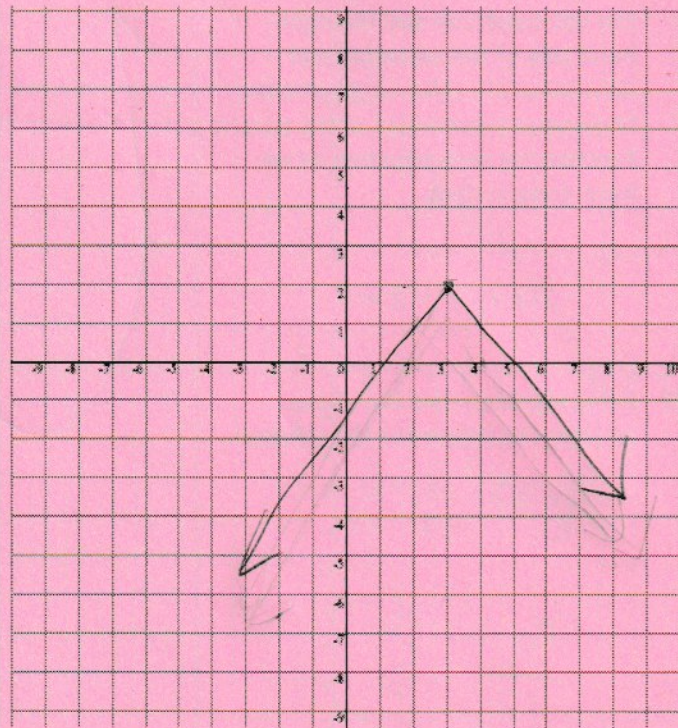
$$6 = 6 \quad \text{or} \quad 70 \neq 6$$

- 7) Using the graph to the right, sketch the left-hand side and the right-hand side of this equation and determine the solution graphically. Show your movements of the parent function for each equation as we have done in class.

$$-|x - 3| + 2 = 1$$

- 8) Using algebra, solve the above equation to determine the solution.

$$\begin{aligned} -|x-3| + 2 &= 1 \\ -|x-3| &= -1 \\ |x-3| &= 1 \\ x-3 &= 1 \quad \text{or} \quad x-3 = -1 \\ x &= 4 \quad \text{or} \quad x = 2 \end{aligned}$$

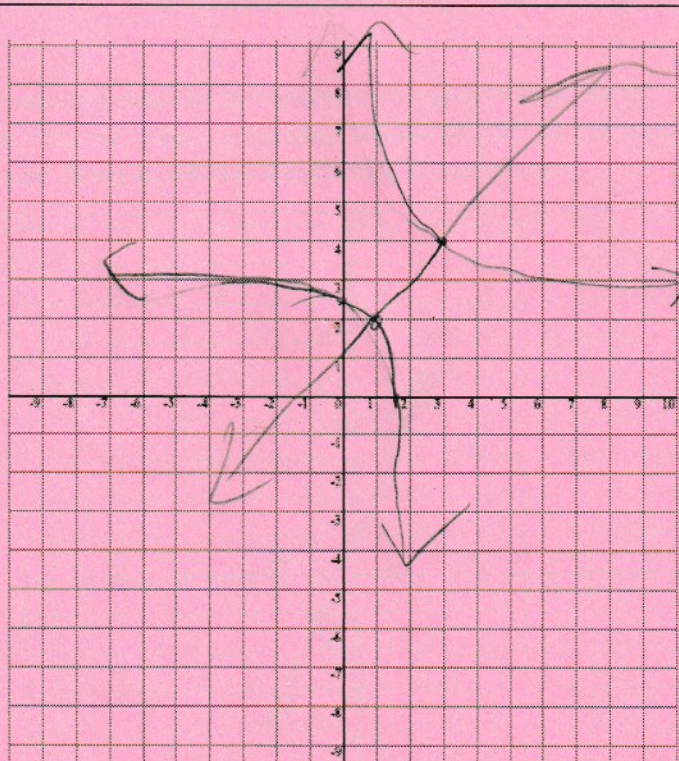


$$\begin{aligned} -|4-3| + 2 &= 1 \quad \text{or} \quad -|2-3| + 2 = 1 \\ -|4-3| &= -1 \quad \text{or} \quad -|2-3| = -1 \\ |4-3| &= 1 \quad \text{or} \quad |2-3| = 1 \\ |1| &= 1 \quad \text{or} \quad |-1| = 1 \\ 1 &= 1 \quad \text{or} \quad 1 = 1 \end{aligned}$$

- 9) Using the graph to the right, sketch the left-hand side and the right-hand side of this equation and determine the solution. Show your movements of the parent function for each equation as we have done in class. Don't forget to draw in the asymptotes.

$$\frac{1}{x-2} + 3 = x + 1$$

- 10) Solve the above equation analytically to determine the solution.



$$\begin{aligned} & \left(\frac{1}{x-2} + 3 \right) = x + 1 \\ & \left(\frac{1}{x-2} \right) = (x-2)(x-2) \\ & x-2 = (x^2-4) \\ & x-2 = 0 \quad \text{or} \quad x^2-4=0 \\ & \textcircled{x=2} \quad \text{or} \quad \textcircled{x=2} \end{aligned}$$

$$\begin{aligned} & \frac{1}{x-2} + 3 = x + 1 \\ & \frac{1}{x-2} = x-2 \\ & \frac{1}{0} = 3 \end{aligned}$$